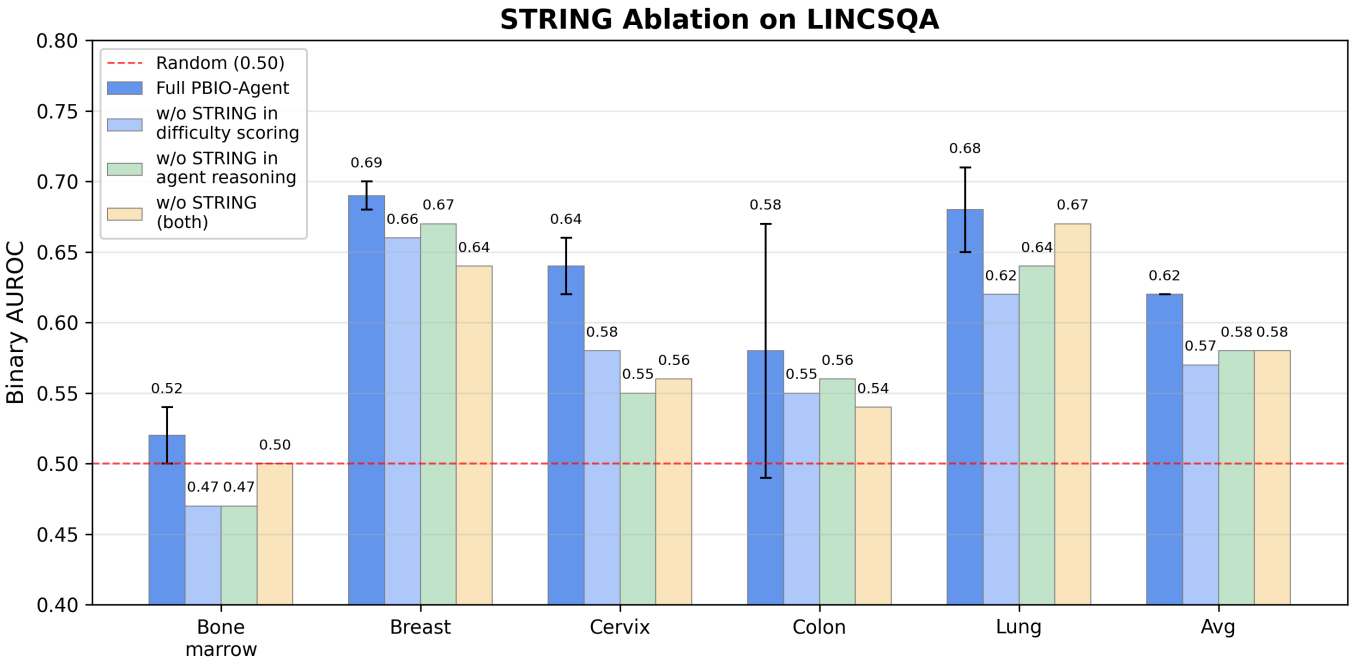

Figure 1: **STRING ablation on LINCSQA.** Binary AUROC scores with STRING components removed from difficulty scoring, agent reasoning, or both. PBIO-Agent retains substantial above-random performance even without STRING, confirming that protein interaction knowledge contributes meaningfully but is not the sole driver of performance.

Judge	Prob. Rate (%)	FPR (%)	FNR (%)	Precision	Recall
History Leakage	3.9	3.7	95.6	0.44	0.04
Target Grounding	4.1	4.4	96.5	0.34	0.03
Answer Consistency	2.0	1.7	97.7	0.47	0.02
Logic	4.3	4.0	95.3	0.43	0.05
Combined (all 4)	10.9	11.0	89.2	0.39	0.11

Table 1: **Judge reliability analysis.** FPR = false positive rate (correct reasoning flagged as problematic); FNR = false negative rate (incorrect reasoning passed). Based on N=863 samples across 4 tissue types.

Aspect	PerturbQA	LINCSQA (Ours)
Perturbation type	Genetic (knockouts/knockdowns)	Chemical (compound treatments)
Cell resolution	Single-cell (Perturb-seq)	Bulk-cell (LINCS L1000)
Gene evaluation	Independent per gene	Progressive (easy→hard)
Biological validity	Accuracy only	Accuracy + MoA-level context
Data source	Perturb-seq experiments	LINCS L1000 database

Table 2: **LINCSQA vs. PerturbQA: key differences.**